

# Sound Waves

## Make a Wave Machine!



# What is Sound?

Sound is a vibration that travels  
through air, water, or solids  
to reach our ears!



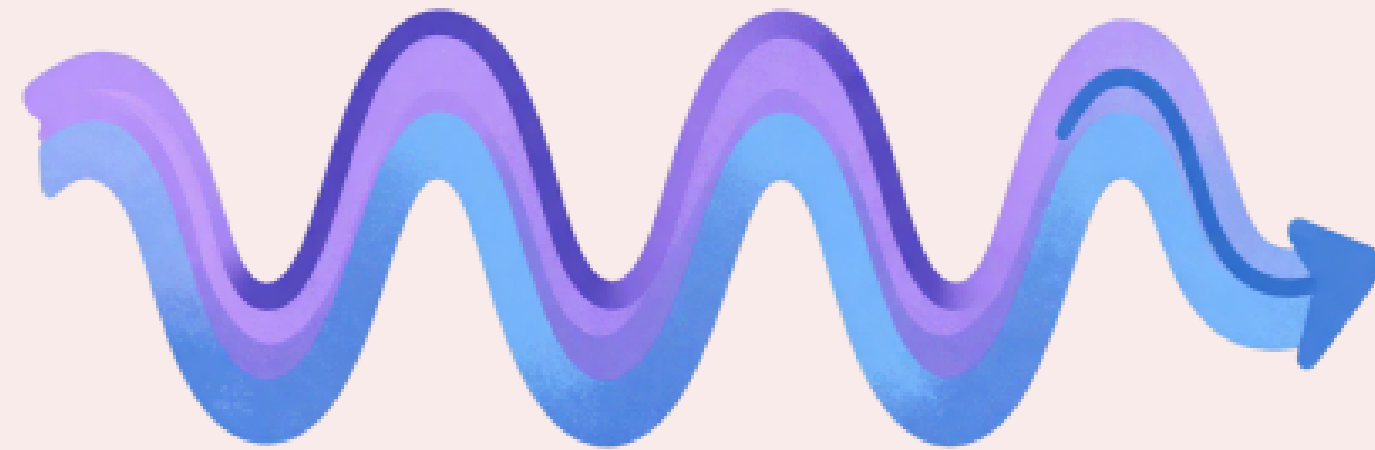
# How Does Sound Travel?

Sound moves in WAVES!  
Just like ripples in a pond  
when you throw a stone.



# What Are Waves?

Waves are patterns of movement  
that carry energy from  
one place to another.

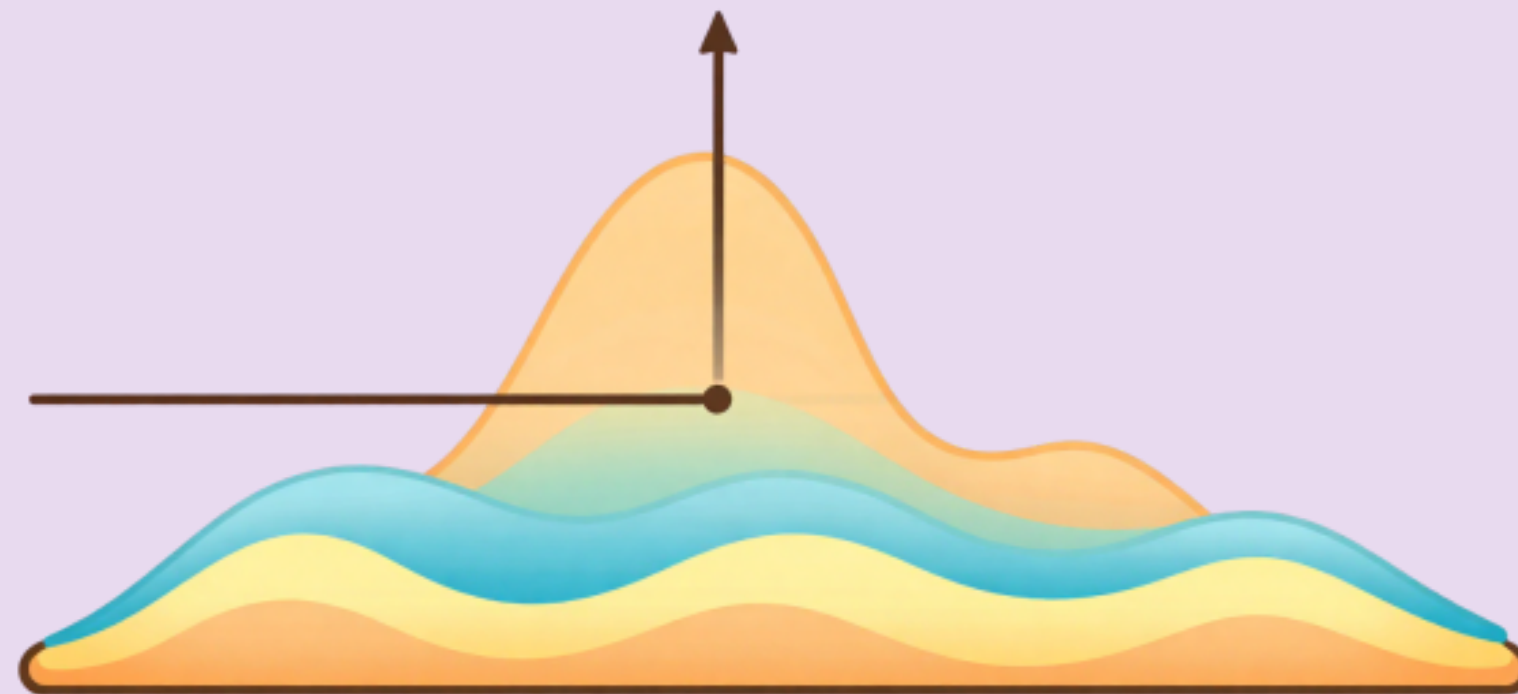


# Parts of a Wave

Crest = the top of a wave

Trough = the bottom of a wave

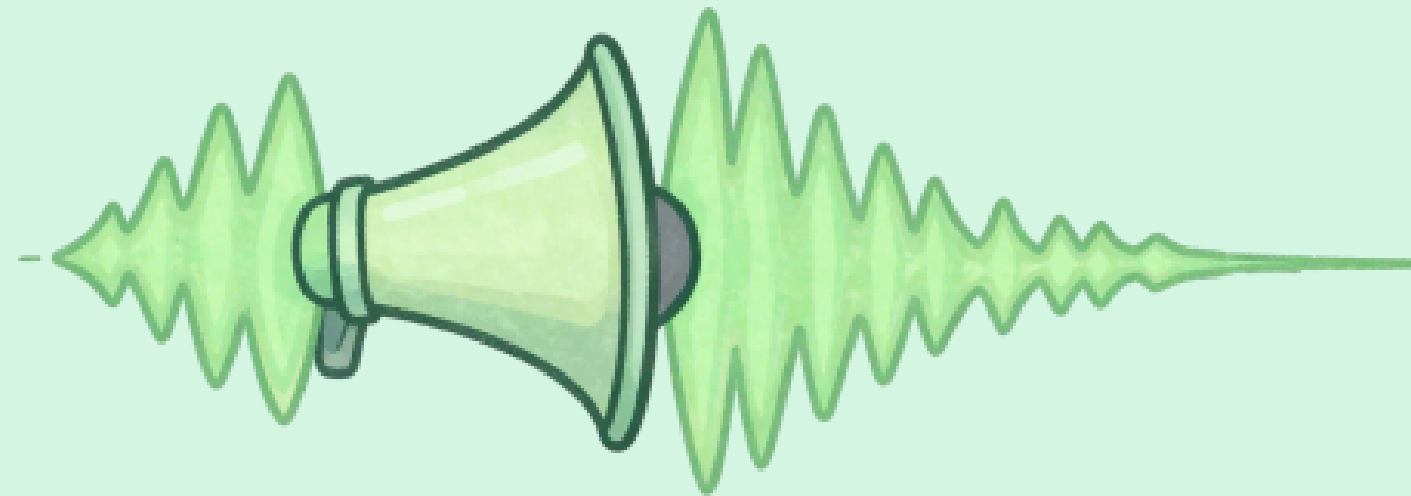
Wavelength = distance between crests



# Loud vs Soft Sounds

Loud sounds = BIG waves (high amplitude)

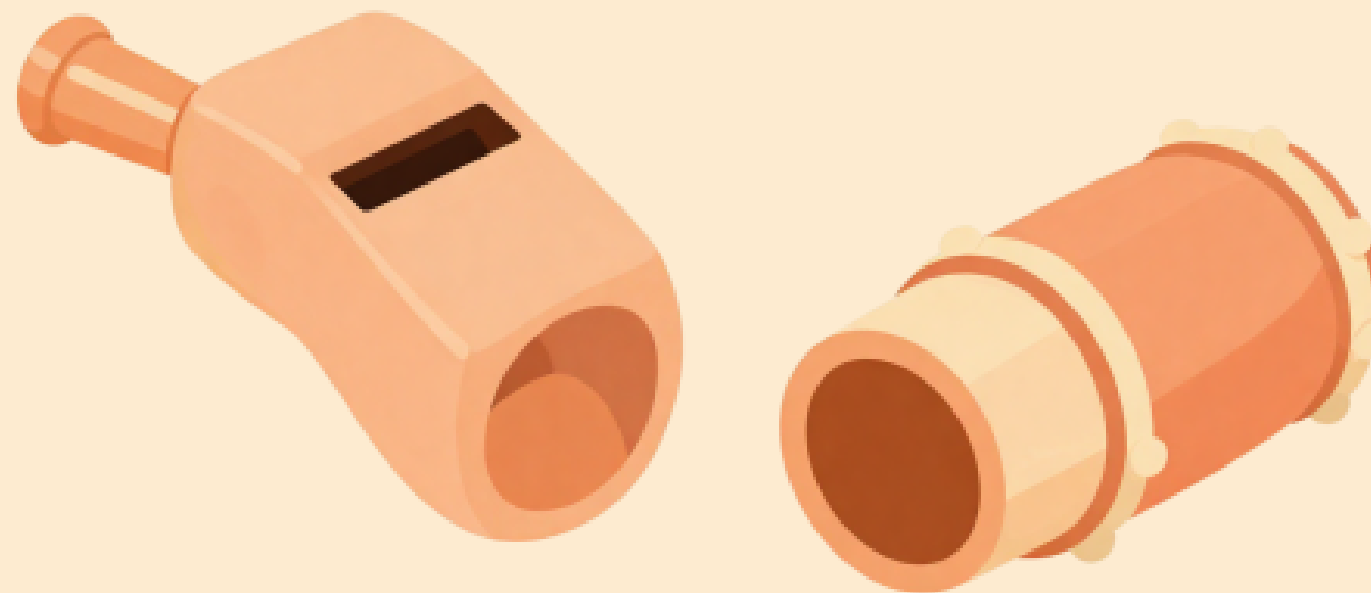
Soft sounds = small waves (low amplitude)



# High vs Low Pitch

High pitch = waves close together (like a whistle)

Low pitch = waves far apart (like a drum)



# Let's Build a Wave Machine!

We will make a simple wave machine  
using everyday materials to SEE  
how sound waves move!



# Materials You Need

30 wooden skewers or sticks

A long strip of duct tape (1 metre)

Small candy or clay balls

A table or flat surface



# Step 1: Lay the Tape

Lay a long strip of duct tape  
sticky side UP on the table.  
Keep it straight and smooth!



## Step 2: Place the Skewers

Place skewers across the tape  
about 2 cm apart from each other.  
Press them down gently.



## Step 3: Add the Balls

Stick small clay balls or candy on BOTH ends of each skewer. These are your wave markers!



## Step 4: Seal It!

Place another strip of tape  
ON TOP to sandwich the skewers.  
Now pick it up carefully!



# Step 5: *Make Waves!*

Hold one end steady.  
Tap a skewer on the other end.  
Watch the wave travel across!



# What Did You Observe?

The wave moves from one end to the other!

Energy transfers through each skewer  
just like sound through air!



# Great Job, Scientists!

You built a wave machine!  
Now you know how sound  
travels as waves of energy!

